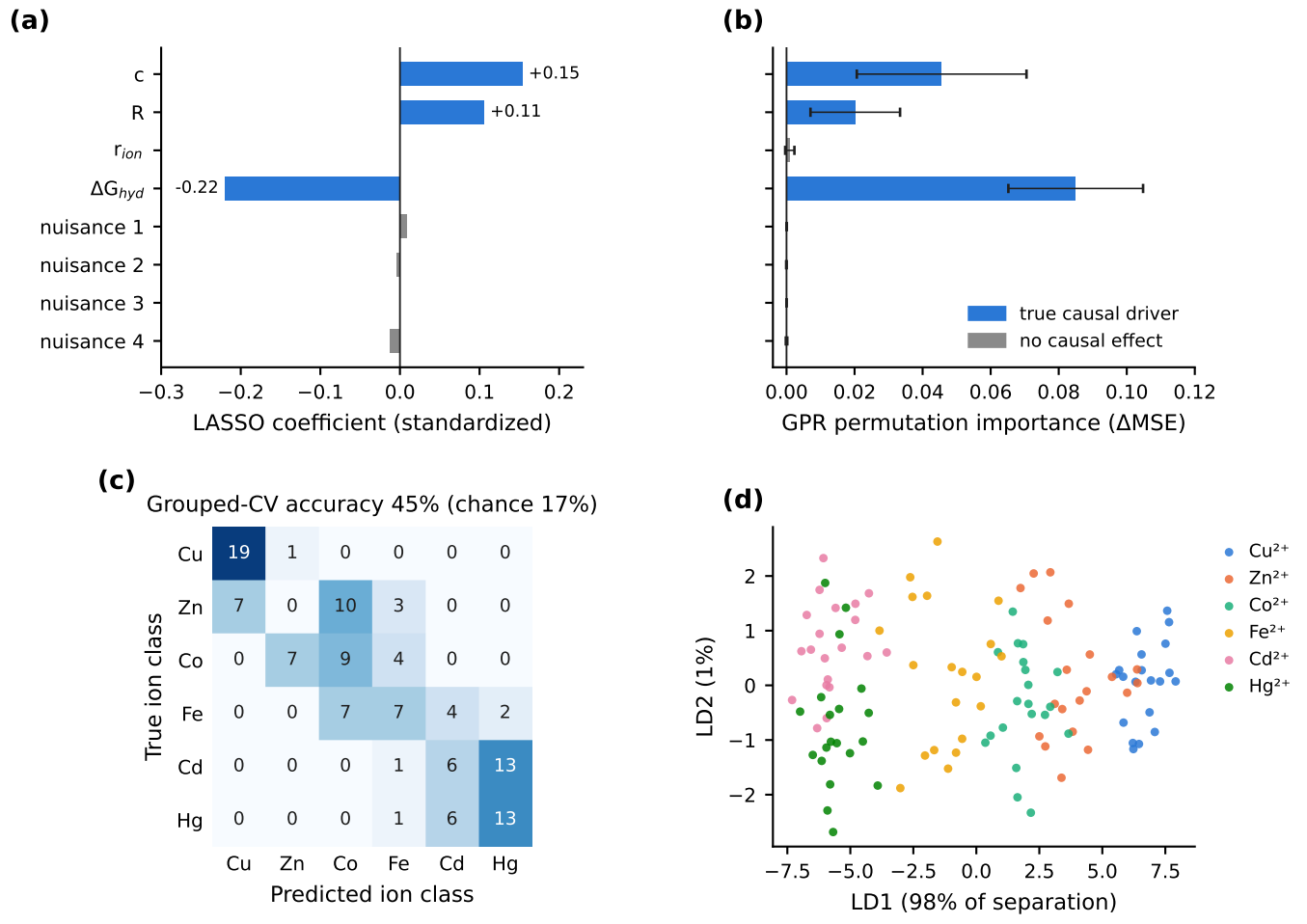
(a)
c
R
r_ion
ΔG_hyd
nuisance 1
nuisance 2
nuisance 3
nuisance 4
+0.15
+0.11
-0.22
−0.3 −0.2 −0.1 0.0 0.1 0.2
LASSO coefficient (standardized)
(b)
true causal driver
no causal effect
0.00 0.02 0.04 0.06 0.08 0.10 0.12
GPR permutation importance (ΔMSE)
(c)
Grouped-CV accuracy 45% (chance 17%)
True ion class
Cu 19 1 0 0 0 0
Zn 7 0 10 3 0 0
Co 0 7 9 4 0 0
Fe 0 0 7 7 4 2
Cd 0 0 0 1 6 13
Hg 0 0 0 1 6 13
Cu Zn Co Fe Cd Hg
Predicted ion class
(d)
LD2 (1%)
2 1 0 −1 −2
−7.5 −5.0 −2.5 0.0 2.5 5.0 7.5
LD1 (98% of separation)
Cu²⁺
Zn²⁺
Co²⁺
Fe²⁺
Cd²⁺
Hg²⁺